



SmartShelf: Edge-Cloud AI for Zero-Waste Kitchens

SmartShelf is an intelligent food management system designed to tackle household food waste—a major contributor to global methane emissions. By combining low-cost IoT hardware with a MERN-stack dashboard and **Agentic AI**, SmartShelf predicts food spoilage in real-time and provides actionable solutions to ensure no ingredient goes to waste.



Key Features

- **Dynamic Expiry Tracking:** Unlike static "best before" dates, SmartShelf uses environmental data to adjust expiry dates based on real-time storage conditions.
- **Agentic AI Assistant ("Shelfy"):** Powered by **Gemini 1.5 Flash**, the built-in chatbot analyzes your inventory to suggest recipes and automatically manages your shopping list.
- **Real-Time Edge-Cloud Sync:** Data entered via the physical 4x4 keypad in your kitchen is instantly reflected on the web dashboard via an ESP32 gateway.
- **Context-Aware Recommendations:** Integrates the **OpenWeatherMap API** to suggest meals based on both your inventory and the local weather (e.g., suggesting soup on a cold day).
- **Visual & Audio Alerts:** Uses an RGB LED and Piezo buzzer to provide immediate freshness feedback without needing to check an app.



System Architecture

The project follows a robust four-layer architecture:

1. **Sensing Layer (Edge Node):** ESP32 microcontroller gathering data from DHT22 (Temperature/Humidity) and DS3231 RTC modules.
2. **Communication Layer:** Bidirectional data transfer using RESTful APIs and BSON serialization via the ArduinoJson library.

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3. **Cloud & Database Layer:** A Node.js backend using a non-blocking I/O model paired with a MongoDB "Digital Pantry".
4. **Application & Intelligence Layer:** A React-based dashboard hosted on Vercel, integrated with Gemini AI for intelligent decision-making.

The Science: Real-Time Spoilage Prediction

SmartShelf moves beyond manual lists by applying food science principles. It utilizes the **Arrhenius Equation** and the **Q_{10} coefficient** to model how chemical reactions (spoilage) accelerate with temperature.

A common rule of thumb in food science is the Q_{10} coefficient, which states that food spoilage typically doubles for every 10°C increase in temperature.

If the sensors detect a rise above the threshold (32.0°C), the system applies a "Virtual Aging" offset, proactively moving the expiry date forward.

Tech Stack

Software

- **Frontend:** React.js, Material-UI (MUI)
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **AI:** Google Gemini 1.5 Flash API
- **Deployment:** Vercel (Frontend), Render (Backend)

Hardware (Simulated in Wokwi)

- **Microcontroller:** ESP32
- **Sensors:** DHT22 (Temp/Humidity), DS1307 RTC
- **I/O:** 16x2 LCD, 4x4 Matrix Keypad, RGB LED, Piezo Buzzer

Future Roadmap: TinyML Integration

The next phase of SmartShelf involves implementing **TinyML** directly on the ESP32. By running lightweight machine learning models locally, the system will be able to:

- Detect spoilage anomalies without internet dependency.
- Process local voice commands.
- Enhance user privacy by keeping data processing at the edge.

Links & Demos

- **Live Dashboard:** <https://smart-shelf-project.vercel.app/>
- **Wokwi Simulation:** <https://wokwi.com/projects/443976664900748289>